

WS & SOA (IT644)

Software Requirements Specification (SRS) / Business Requirements Document (BRD)

PrepNova

RAG-based Adaptive Mock Interview Platform

GitHub Repository: github.com/Krishna4801/PrepNova

Project Team

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1. Project Overview

PrepNova is an AI-powered mock interview platform designed to help students and job seekers prepare for technical and HR interviews. Instead of practicing from static question banks, users upload their resume or choose a target role, and the platform generates personalized interview questions using a Retrieval-Augmented Generation (RAG) pipeline combined with the open source Api.

After completing an interview, PrepNova evaluates responses and provides an ATS score, technical assessment, communication analysis, strengths, weaknesses, and actionable feedback. The platform also tracks interview history and learning progress through analytics.

Target Users

- College students preparing for placements
- Freshers applying for software roles
- Professionals switching careers
- Candidates preparing for HR and technical interviews

2. Problem Statement

Many candidates struggle to prepare for interviews because available platforms either provide generic questions or lack personalized feedback. Recruiters increasingly evaluate resumes, communication skills, and practical knowledge together. PrepNova addresses this with:

- Resume-based personalized interviews
- Role-specific interview generation
- AI evaluation and ATS scoring
- Performance analytics and progress tracking
- Interview history with detailed reports

3. Objectives

Primary Objectives

- Generate AI-powered interview questions
- Support resume and role-based interviews
- Evaluate candidate responses intelligently
- Provide ATS and confidence scores
- Track user progress over time

Secondary Objectives

- Improve interview readiness
- Identify skill gaps
- Build confidence through repeated practice
- Create a clean SaaS learning experience

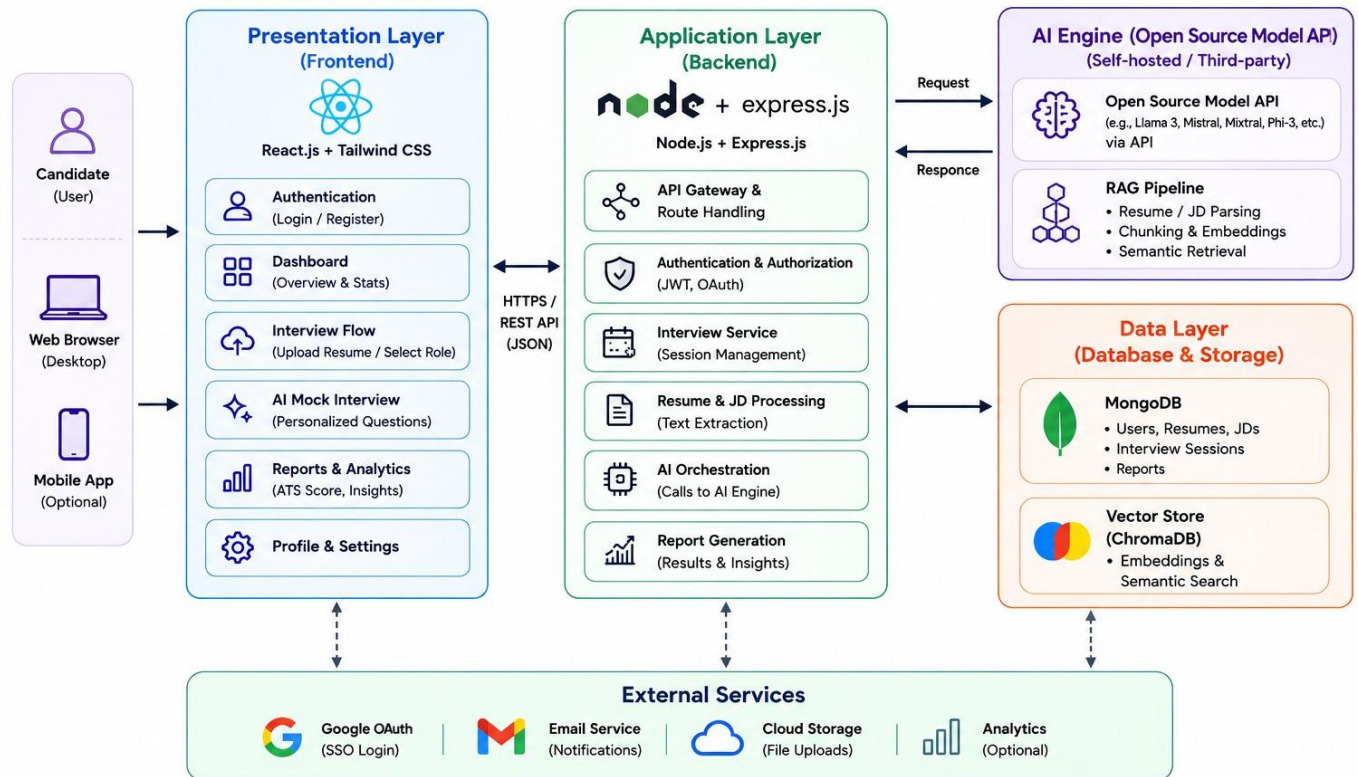
4. Technology Stack

Layer	Technology
Frontend	React.js + Tailwind CSS
Backend (API)	Node.js + Express.js
AI Microservice	Python + FastAPI (hosts the RAG pipeline)
Database	MongoDB
Vector Store	chromaDB (chunk embeddings for RAG retrieval)
AI / LLM Engine	open source model api
File Storage	Cloudinary
Routing	React Router DOM
Charts	Recharts
Authentication	JWT (httpOnly cookie) + Google OAuth — avoid storing JWT in localStorage (XSS risk)

5. System Architecture

PrepNova – System Architecture

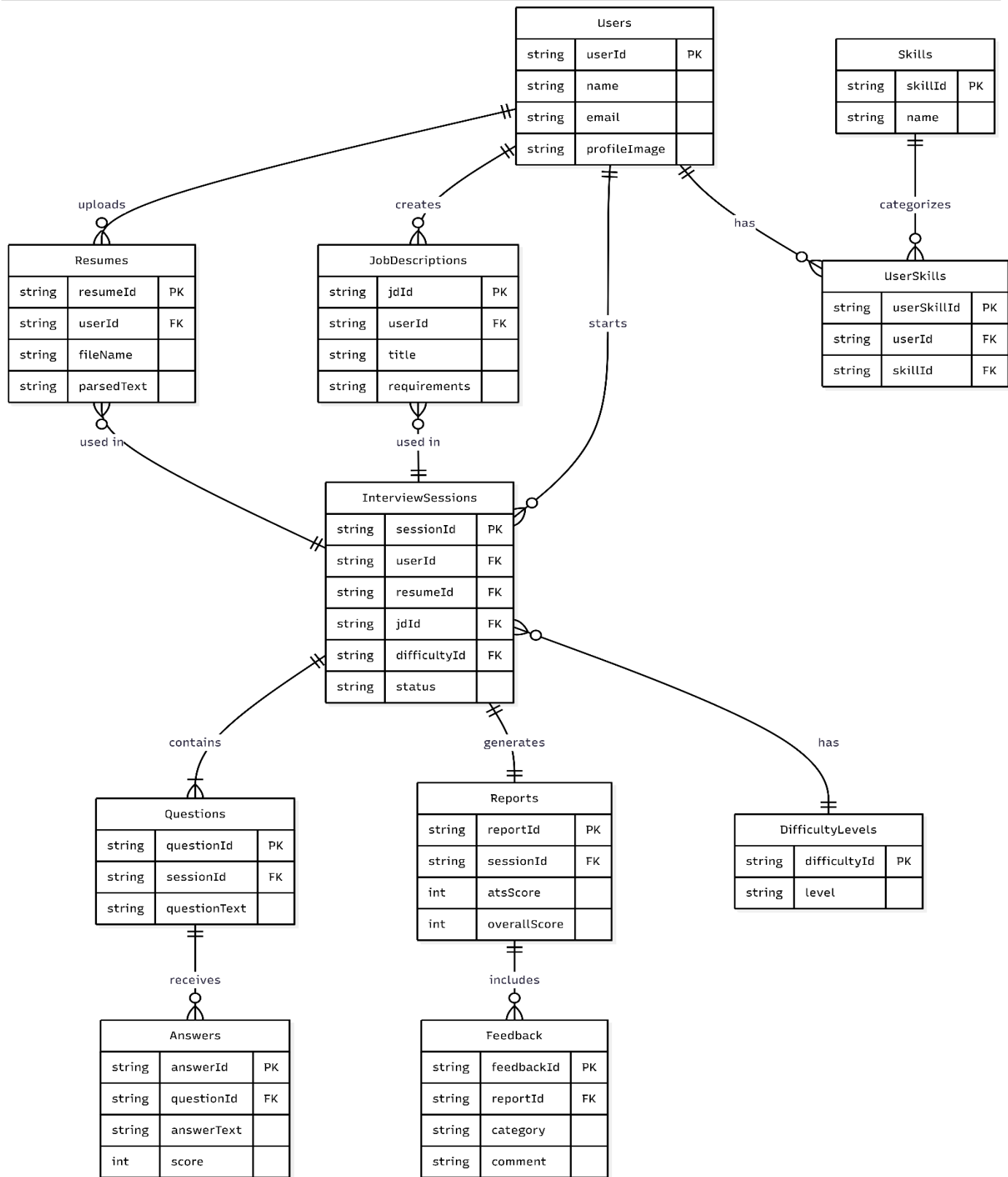
AI-Powered Mock Interview Platform



Data Flow

- 1 User uploads resume.
- 2 Backend extracts resume text and forwards it to the AI microservice.
- 3 AI microservice chunks the text, generates embeddings, and stores them in the vector store.
- 4 On interview start, the RAG retriever fetches the most relevant chunks and builds a prompt for Open source Api.
- 5 Open source Api generates personalized questions.
- 6 User answers questions.
- 7 AI evaluates responses.
- 8 Report is stored in MongoDB.
- 9 Dashboard displays analytics.

6. Database Schema — ER Diagram

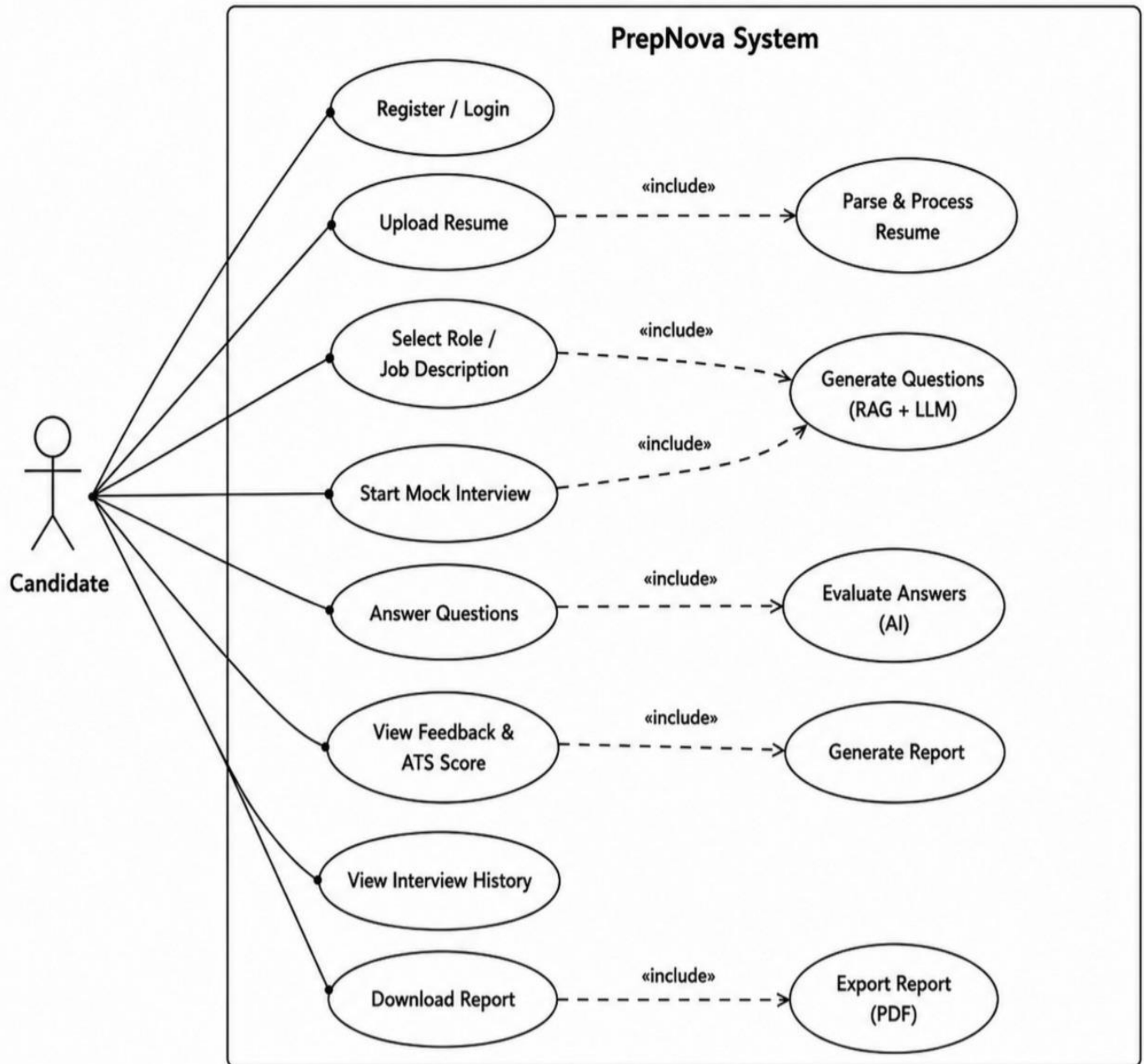


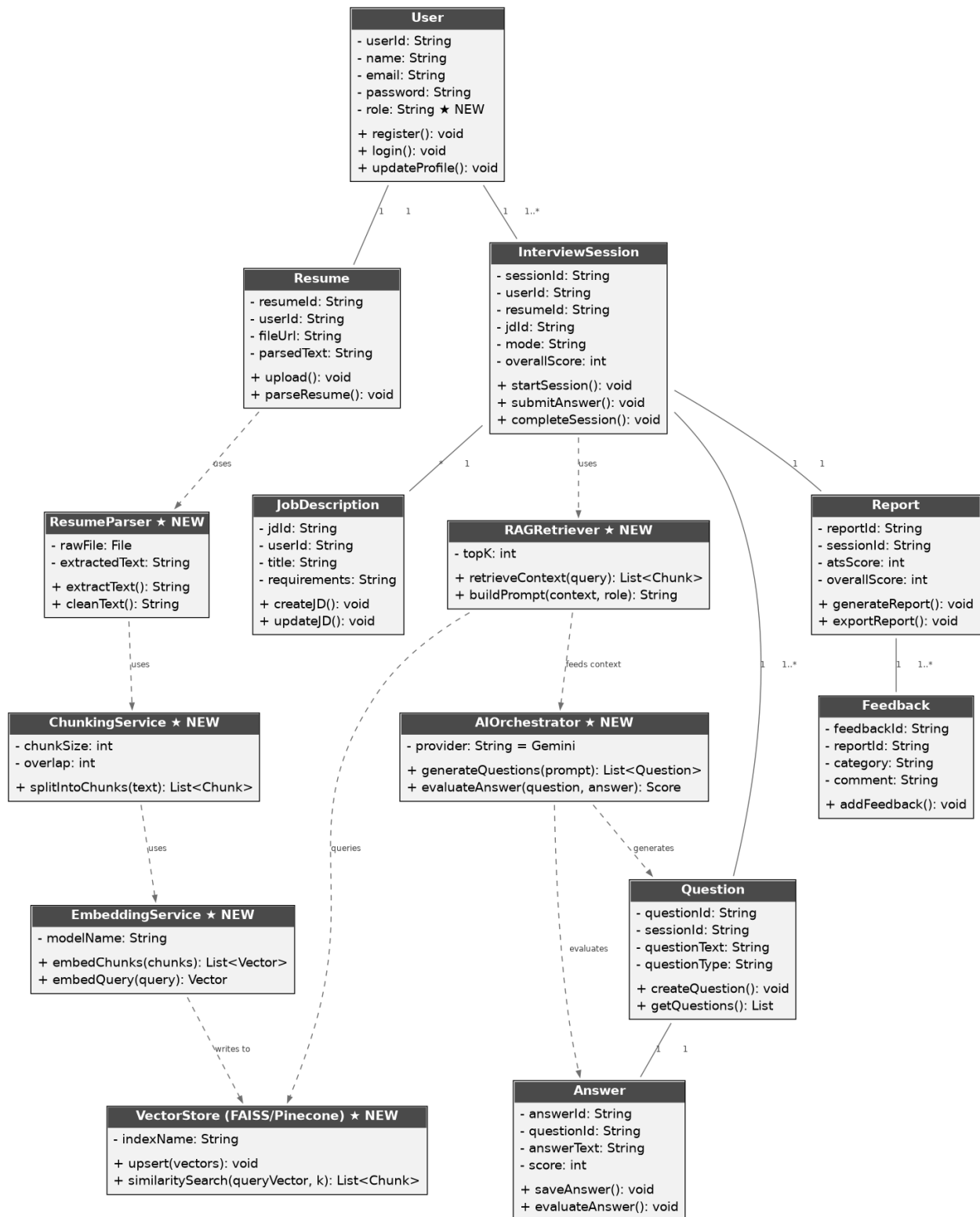
Entity Description

Entity	Description
Resume	Uploaded resume and parsed text.
User	Stores candidate profiles, credentials, and account metadata
JobDescription	Optional job description used for personalization.
InterviewSession	Complete interview metadata; liveScore updates as the session progresses.
Question	AI-generated interview questions.
Answer	User responses and per-answer scores.
Report	ATS score and final, authoritative overall evaluation.
Feedback	Strengths, weaknesses and suggestions.
Skill / UserSkill	Master skill catalogue and candidate skill proficiency.
DifficultyLevel	Easy, Medium and Hard interview levels.

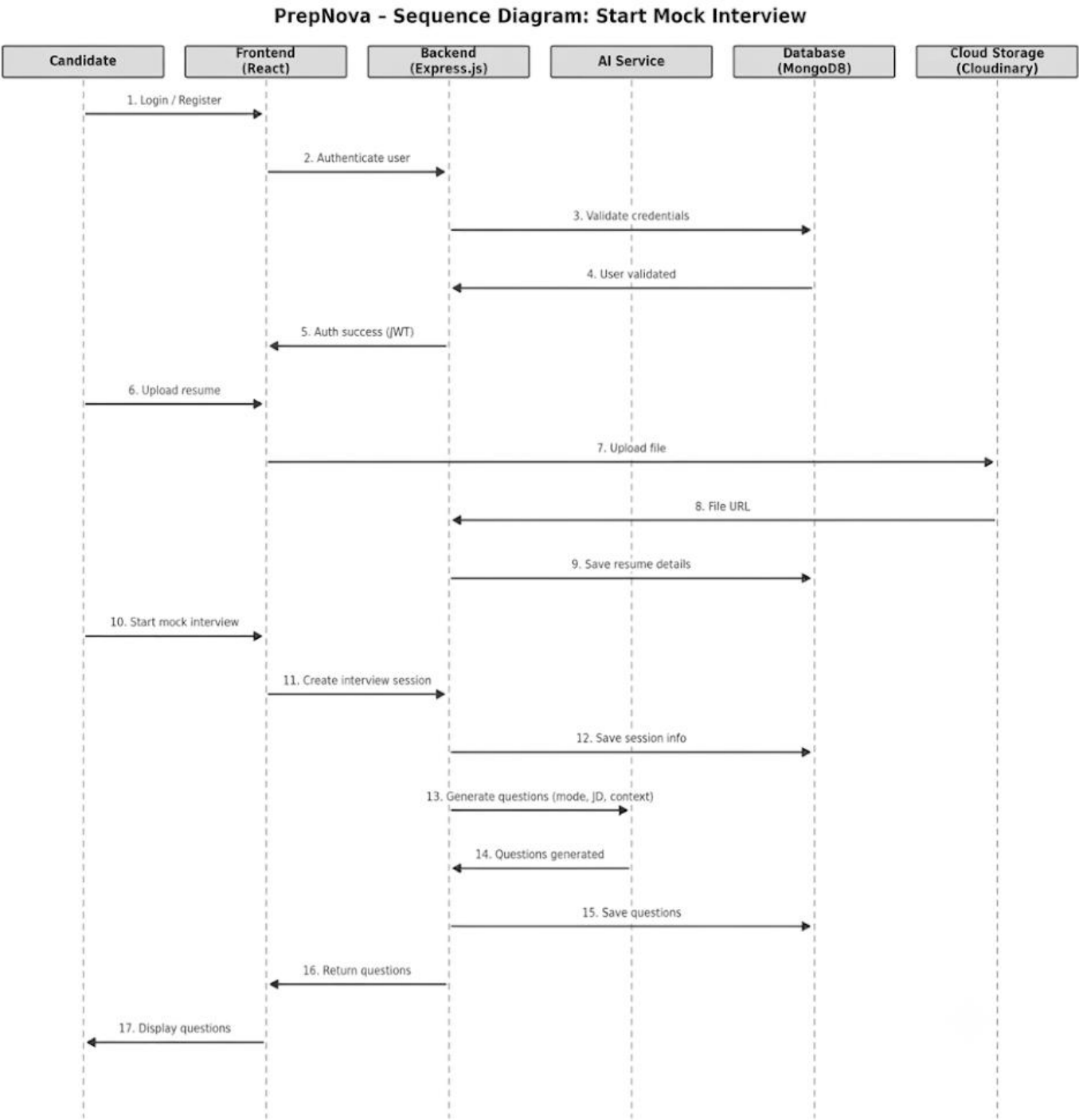
7. UML Diagrams

7.1 Use Case Diagram



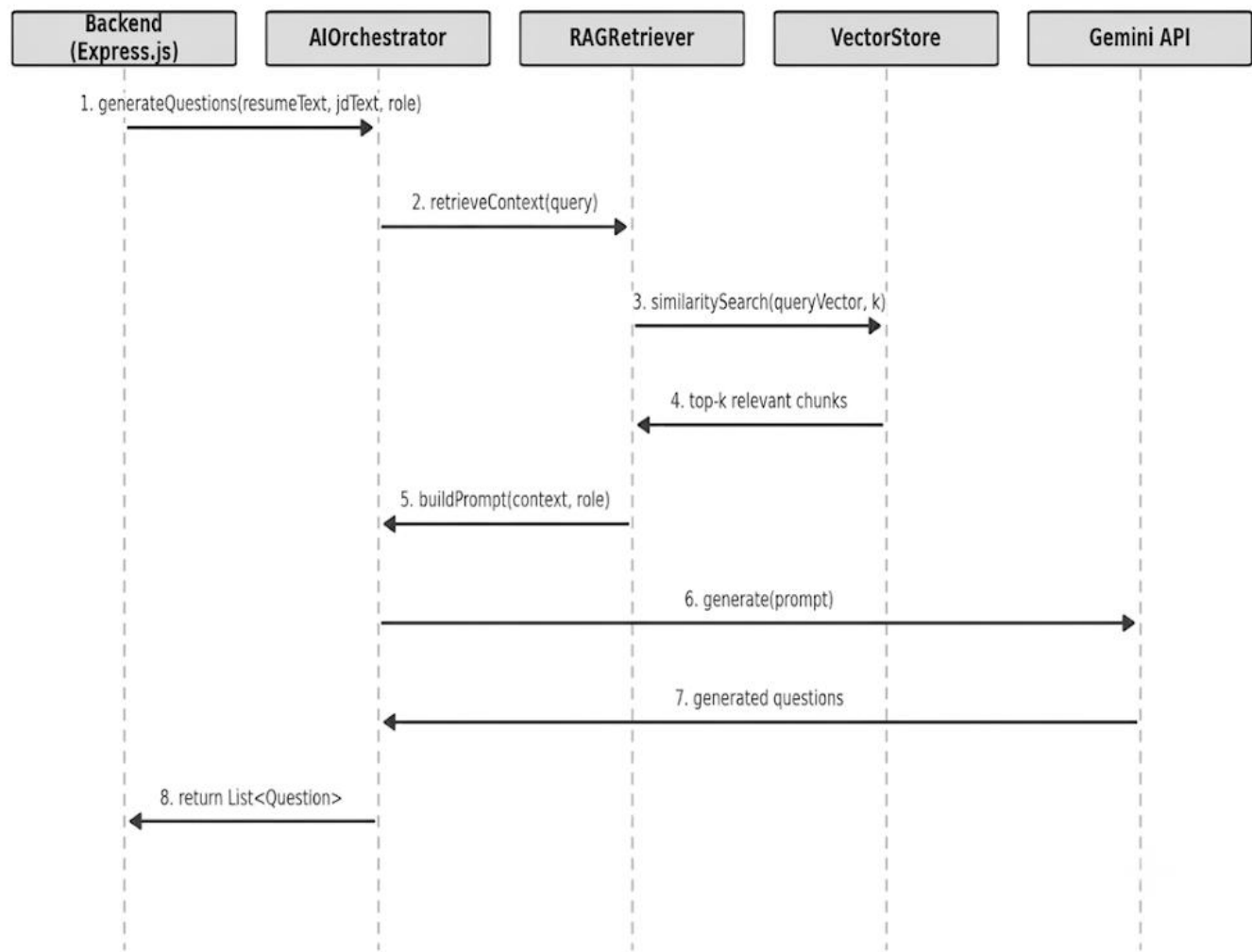


7.3 Sequence Diagram — Start Mock Interview



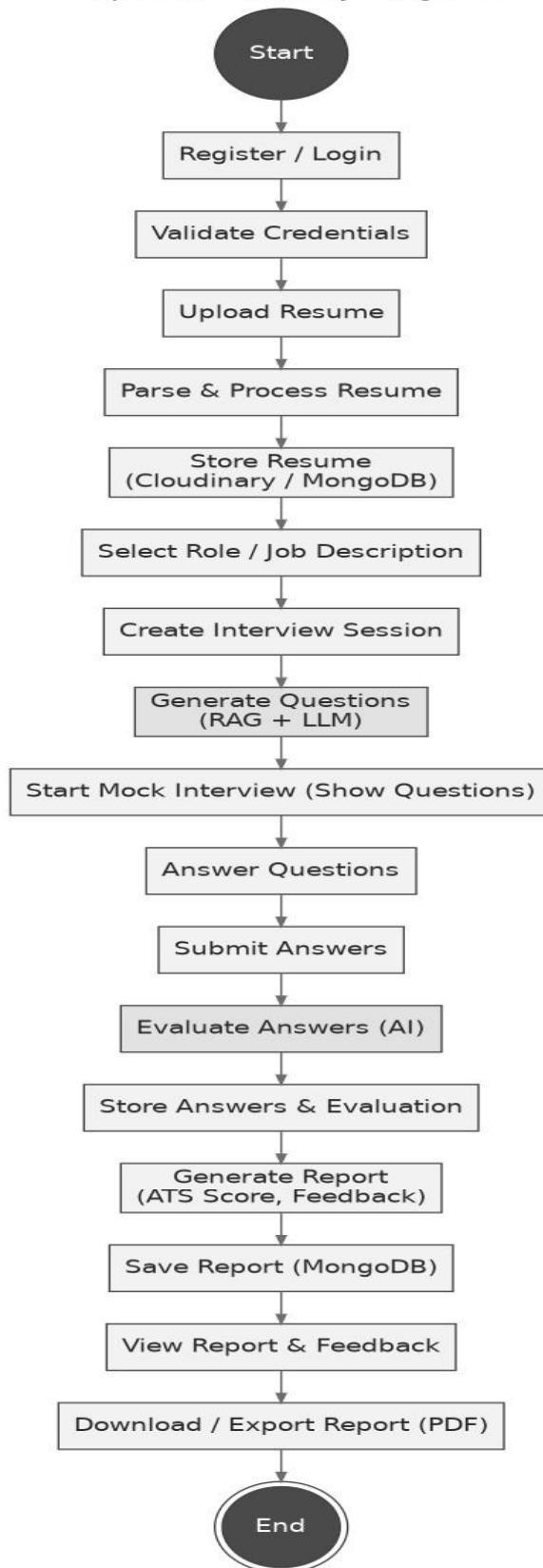
7.4 Sequence Diagram — RAG-based Question Generation

PrepNova - Sequence Diagram: RAG-based Question Generation (NEW)



7.5 Activity Diagram

PrepNova – Activity Diagram



8. REST API Contracts

Expanded from 6 endpoints to full CRUD coverage of every entity in the ER diagram, with status codes and error responses on each endpoint — both missing from the original. Base path: `/api`.

8.1 Authentication

POST	<code>/auth/signup</code>
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Purpose: Create a new account.

Request: { name, email, password } Responses: 201 Created { token, user } | 400 Bad Request (validation) | 409 Conflict (email exists)

POST	<code>/auth/login</code>
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Purpose: Authenticate a user.

Request: { email, password } Responses: 200 OK { token, user } | 401 Unauthorized (bad credentials)

POST	<code>/auth/oauth/google</code>
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Purpose: Google OAuth SSO login.

Request: { idToken }
Responses: 200 OK { token, user } | 401 Unauthorized

GET	<code>/auth/me</code>
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Purpose: Return the current authenticated user's profile.

Responses: 200 OK { user } | 401 Unauthorized

8.2 Resume & Job Description Module

POST	<code>/resume/upload</code>
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Purpose: Upload a PDF resume (multipart/form-data).

Request: multipart/form-data: file Responses: 201 Created { resumeId, status } | 415 Unsupported Media Type

GET	<code>/resume/:resumeId</code>
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Purpose: Fetch a resume's parsed text and metadata. Responses:

200 OK { resume } | 404 Not Found

DELETE	<code>/resume/:resumeId</code>
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Purpose: Delete an uploaded resume.

Responses: 204 No Content | 404 Not Found

POST	<code>/jd</code>
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Purpose: Create a job description for role-based interviews.

Request: { title, company, requirements }
Responses: 201 Created { jdId } | 400 Bad Request

GET	<code>/jd/:jdId</code>
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Purpose: Fetch a job description.

Responses: 200 OK { jd } | 404 Not Found

PUT /jd/:jdId

Purpose: Update a job description.

Request: { title?, company?, requirements? }

Responses: 200 OK { jd } | 404 Not Found

DELETE /jd/:jdId

Purpose: Delete a job description.

Responses: 204 No Content | 404 Not Found

Interview Module

POST /interview/start

Purpose: Create a new interview session (triggers RAG question generation).

Request: { mode, difficulty, resumeId?, jdId? } Responses: 201 Created { sessionId, questions[] } | 400 Bad Request | 502 Bad Gateway (AI service unreachable)

POST /interview/answer

Purpose: Submit one interview answer for AI evaluation.

Request: { sessionId, questionId, answer } Responses: 200 OK { score, feedback } | 404 Not Found (session/question) | 422 Unprocessable Entity

POST /interview/:sessionId/complete

Purpose: Mark a session complete and trigger report generation.

Responses: 200 OK { reportId } | 404 Not Found | 409 Conflict (already completed)

GET /interview/:sessionId

Purpose: Fetch session details, questions, and answers so far.

Responses: 200 OK { session } | 404 Not Found

8.3 Report Module

GET /report/:sessionId

Purpose: Return the complete ATS report for a session.

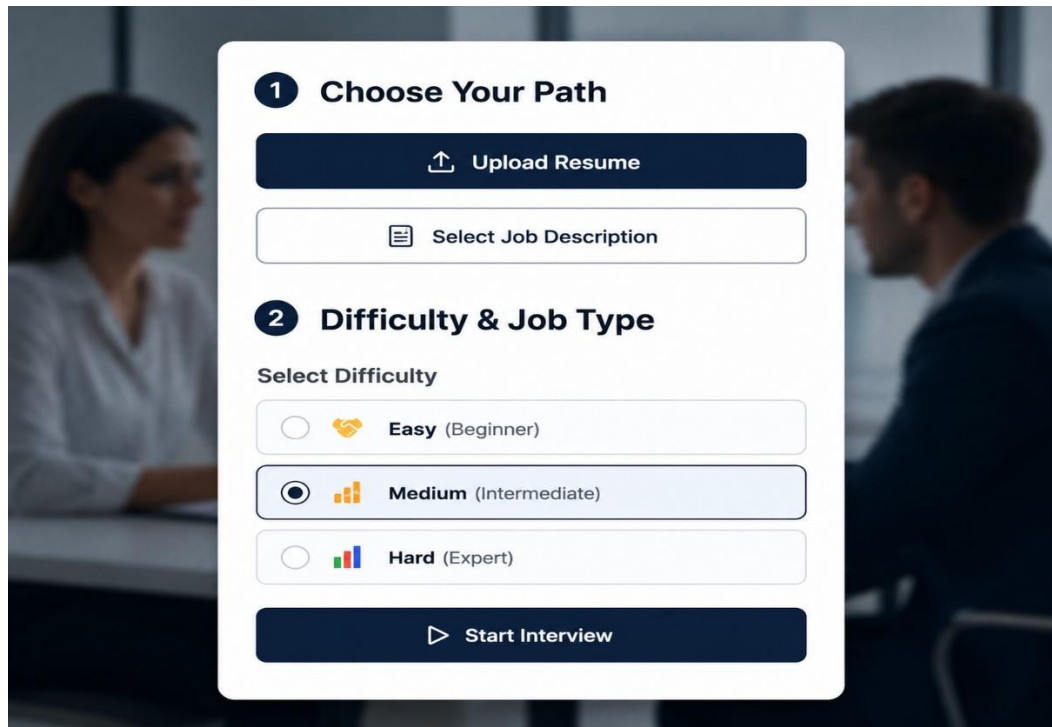
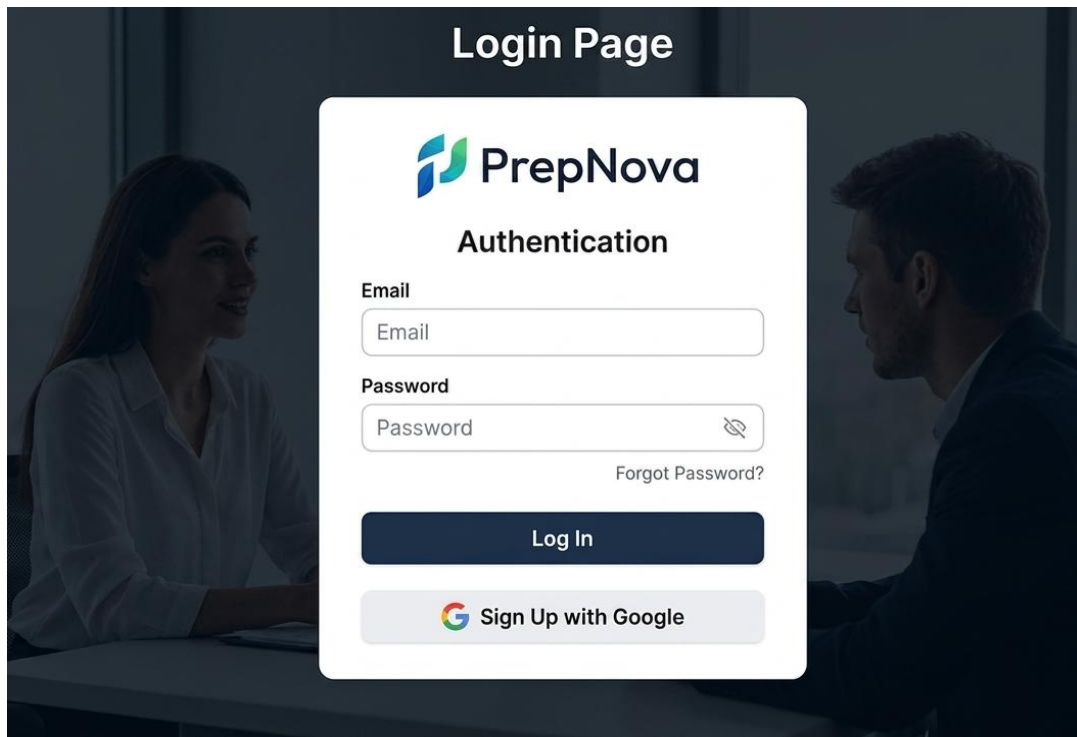
Responses: 200 OK { atsScore, overallScore, confidence, strengths[], weaknesses[], suggestions[] } | 404 Not Found

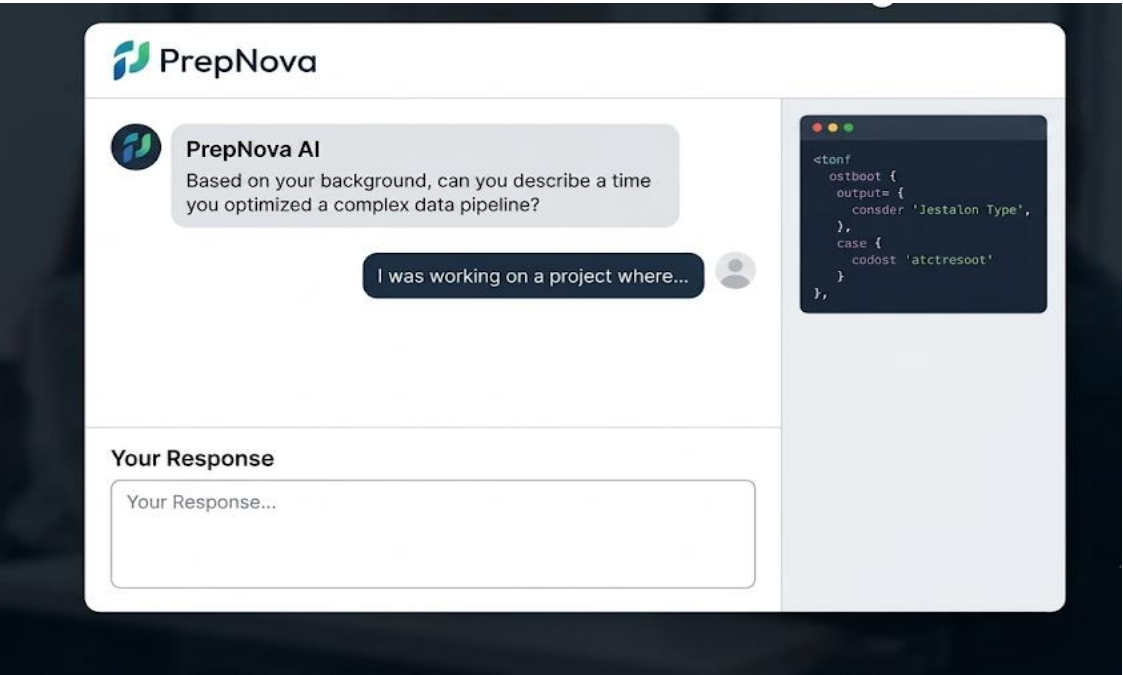
GET /report/:sessionId/export

Purpose: Export the report as a downloadable PDF.

Responses: 200 OK (application/pdf) | 404 Not Found

9. UI/UX Mockups





10. Future Enhancements

- Voice-based AI interviews
- Real-time coding editor
- Video interview analysis
- Company-specific interview datasets
- Recruiter dashboard
- Leaderboard & gamification
- Mobile application